

Course Title :Generative AI

Unit No : III

Topics :

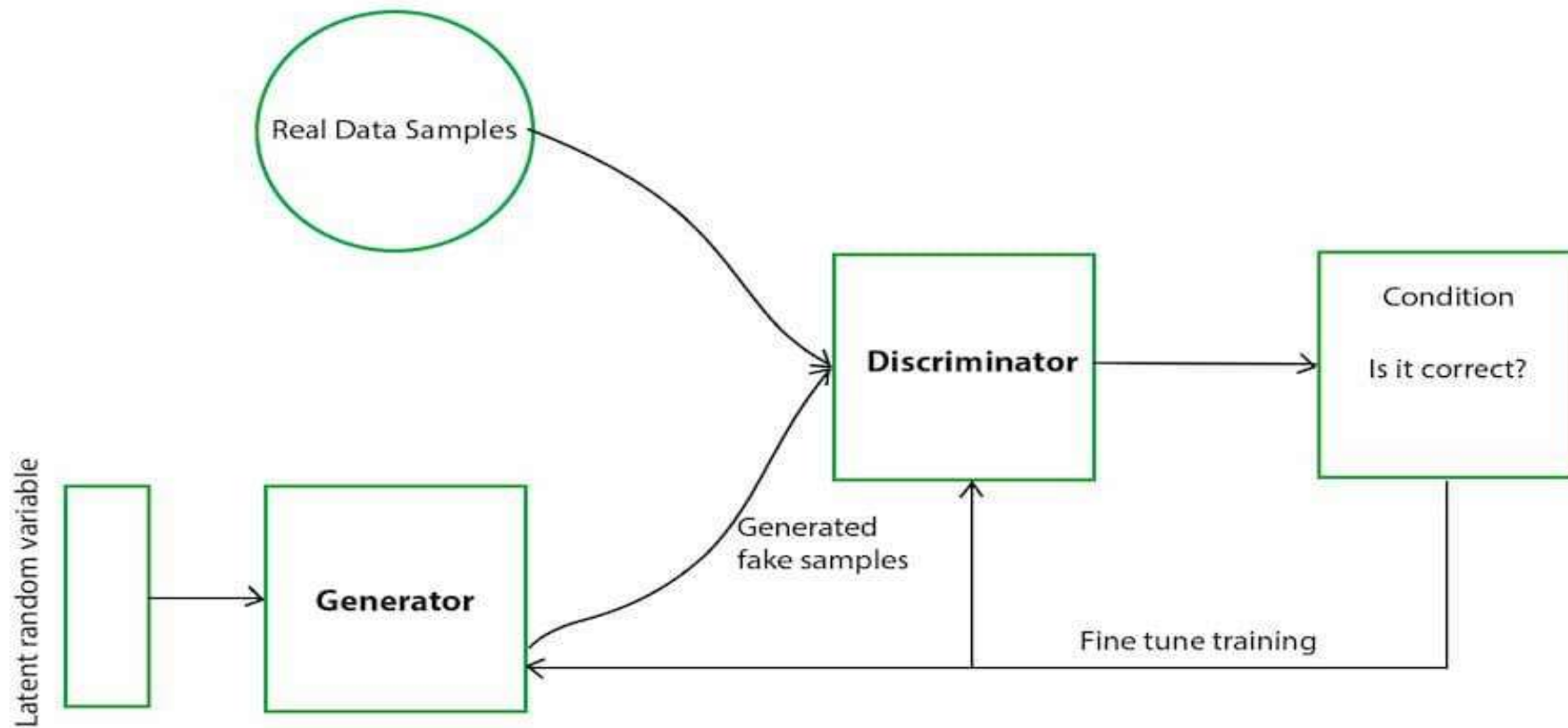
UNIT- III Generation of Images: Introduction to Generative Adversarial Networks, Adversarial Training Process, Nash Equilibrium, Variational Autoencoders, Encoder-Decoder Architectures, Introduction to Transformer-based Image Generation, CLIP, Visual Transformers ViT- Dall-E2 and Dall-E3, GPT-4V, Issues of Image Generation models

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Introduction to Generative Adversarial Networks

- This is the core idea behind Generative Adversarial Networks (GANs), one of the most brilliant and influential concepts in modern artificial intelligence.
- **What is a GAN?**
- A Generative Adversarial Network (GAN) is a class of machine learning frameworks where two neural networks contest with each other in a game (in the sense of game theory, hence "adversarial"). This contest results in the generation of new, synthetic data that can pass for real data.
- The key insight is that instead of having one network learn to generate data on its own, we have two networks:

- **The Generator:** The "forger." It tries to create fake data.
- **The Discriminator:** The "art expert." It tries to distinguish real data from the Generator's fakes.



- The Two Key Players
- **1. The Generator (G)**
- Goal: To generate plausible, fake data. It wants to fool the Discriminator into thinking its outputs are real.
- Input: It takes in a random noise vector (often called a latent vector or seed). This randomness ensures the generator can produce a wide variety of outputs, not just one thing.
- Output: Synthetic data (e.g., an image of a face that doesn't exist).
- Analogy: A counterfeiter printing fake money.

- **2. The Discriminator (D)**
- Goal: To correctly classify data as either "real" (from the actual dataset) or "fake" (from the Generator).
- Input: A piece of data (could be real or generated).
- Output: A probability score between 0 and 1, where 1 means "100% confident it's real" and 0 means "100% confident it's fake."
- **Generative Adversarial Networks represent a paradigm shift in AI, moving from simply *recognizing* patterns to actively *creating* new content. By pitting two networks against each other in a creative arms race, GANs have unlocked a new frontier of generative AI, with applications spanning from art and entertainment to science and medicine.**

Adversarial Training Process

- The training process is a elegant minimax game—a competition where each player's move is a counter to the other's.
- The Discriminator's Turn: A batch of real images from the dataset is mixed with a batch of fake images from the Generator. The Discriminator is trained on this mixed batch. Its weights are updated to get better at telling real from fake. "This is what a real face looks like, and this is what a bad fake looks like."
- The Generator's Turn: The Generator creates a new batch of fakes. This time, we freeze the Discriminator's weights (so it can't learn) and run the fakes through it. The error (how well the fakes fooled the Discriminator) is used to update the Generator's weights. "Okay, my last fakes were detected. Let me adjust to make them more realistic based on what the Discriminator is looking for."

Applications of GANs

- GANs are incredibly versatile and have been used to create astonishing results:
- **Image Generation:** Creating photorealistic images of human faces, animals, and scenes that never existed (e.g., thispersondoesnotexist.com).
- **Image-to-Image Translation:** Converting satellite photos to maps, black-and-white photos to color, sketches to realistic images, day scenes to night.
- **Style Transfer:** Applying the artistic style of one image (e.g., Van Gogh's *Starry Night*) to a photograph.
- **Super-Resolution:** Enhancing the resolution and quality of low-resolution images ("enhance!" from CSI is becoming a reality).
- **Drug Discovery:** Generating new molecular structures for novel medicines.
- **Deepfakes:** Realistically replacing a person in a video with someone else (a significant ethical and societal challenge).

- This two-step process repeats thousands or millions of times. The entire training can be summarized by this objective function:
- **In simple terms:** The Discriminator (D) tries to **maximize** the probability of correctly labeling real and fake data (it wants $D(\text{real})$ to be 1 and $D(\text{fake})$ to be 0). Meanwhile, the Generator (G) tries to **minimize** the probability that the Discriminator is correct (it wants $D(\text{fake})$ to be 1).

Nash Equilibrium

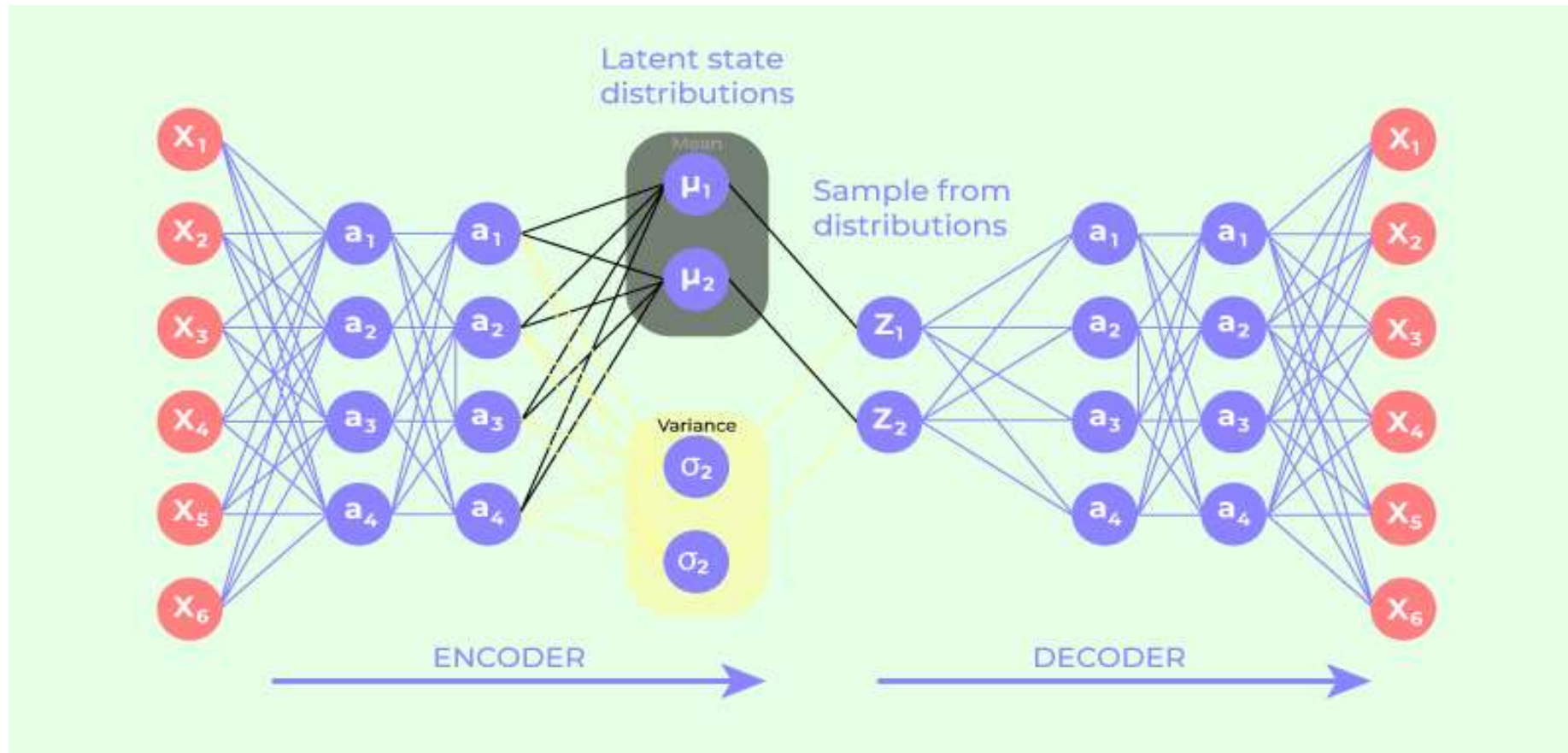
- The ideal end state of this game is a **Nash Equilibrium**, where:
- The Generator produces such perfect fakes that the Discriminator is completely fooled.
- The Discriminator is forced to guess randomly, giving a probability of 0.5 (50/50) to every image, as it can no longer tell the difference.
- At this point, the Generator has learned the true distribution of the training data and can create entirely new samples that are indistinguishable from reality.

Variational Autoencoders

- **Variational Autoencoders (VAEs)** are type of generative model in machine learning that create new data similar to the input they are trained on. They not only compress and reconstruct data like traditional autoencoders but also learn a continuous probabilistic representation of the underlying features.
- This unique approach helps VAEs to generate new, realistic data samples that closely resemble the original input.
- In this article, we will see more about VAEs and its core concepts.
- A **VAE is a probabilistic autoencoder that learns a continuous, smooth latent space**, enabling both **reconstruction** and **generation of new samples**.

- **Architecture of Variational Autoencoder**
- VAE is a special kind of autoencoder that can generate new data instead of just compressing and reconstructing it. It has three main parts:
- **1. Encoder (Understanding the Input)**
- The encoder takes input data like images or text and learns its key features. Instead of outputting one fixed value, it produces two vectors for each feature:
- **Mean (μ):** A central value representing the data.
- **Standard Deviation (σ):** It is a measure of how much the values can vary.

- These two values define a range of possibilities instead of a single number.
- **2. Latent Space (Adding Some Randomness)**
- Instead of encoding the input as one fixed point it pick a random point within the range given by the mean and standard deviation. This randomness lets the model create slightly different versions of data which is useful for generating new, realistic samples.
- **3. Decoder (Reconstructing or Creating New Data)**

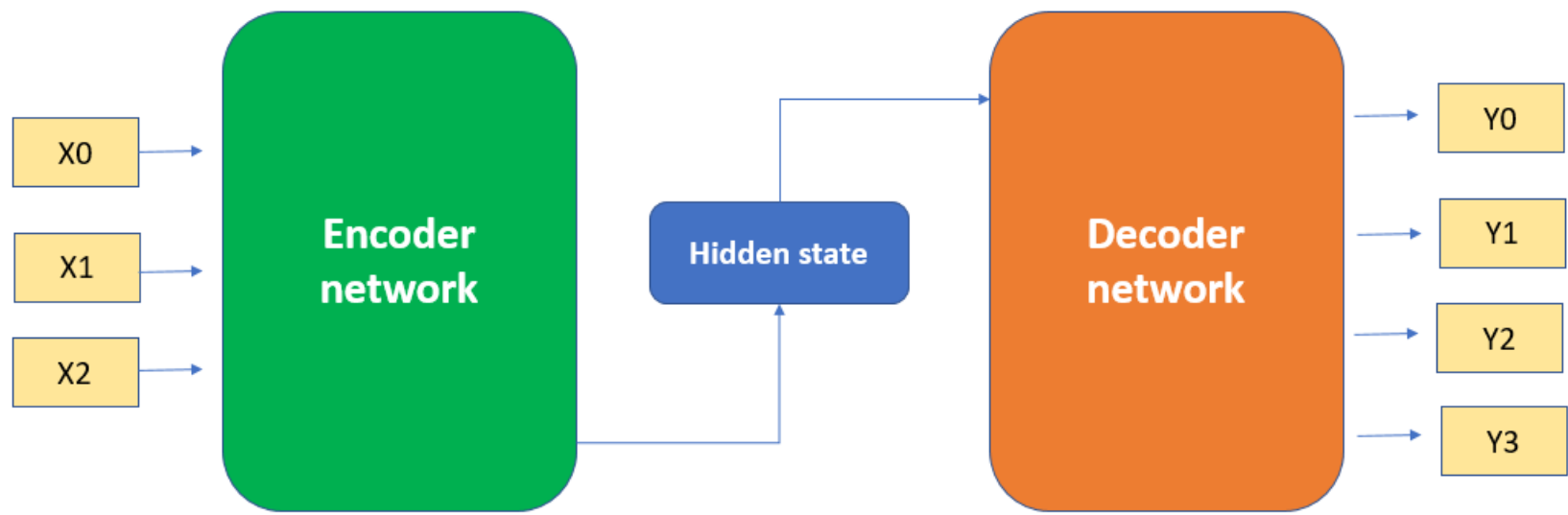


Applications

- **Image generation** (like handwritten digits, faces).
- **Data compression** with a probabilistic model.
- **Representation learning** (latent space captures meaningful features).
- **Anomaly detection** (unusual inputs are poorly reconstructed).
- **Molecular design / drug discovery** (generating new chemical structures).

Encoder-Decoder Architectures

- Encoder-decoder architectures are deep learning frameworks for **sequence-to-sequence tasks**, comprising an encoder that converts an input sequence into a fixed-length vector (context vector) and a decoder that generates an output sequence from this vector.
- Popular applications include **machine translation, text summarization, image captioning, and speech recognition**.
- The encoder processes the entire input, while the decoder produces the output step-by-step, with both components often implemented using architectures like [Recurrent Neural Networks \(RNNs\)](#) or [Transformers](#).



Key Applications

- **Machine Translation**: Translating sentences from one language to another.
- **Text Summarization**: Condensing long articles into shorter summaries.
- **Image Captioning**: Generating descriptive text for an image.
- **Speech Recognition**: Converting spoken language into text.
- **Large Language Models**

Introduction to Transformer-based Image Generation

- **How It Works: The Two-Stage Process**
- Most Transformer-based image generation models follow a similar pattern:
- **Stage 1: Tokenization - From Image to Sequence**
- You cannot feed raw pixels directly into a Transformer. The image must be broken down into a sequence of tokens. There are two primary methods:
- **Patch-based Tokenization (The ViT Approach):**
 - The image is split into a grid of fixed-size patches (e.g., 16x16 pixels).
 - Each patch is flattened into a vector.
 - These vectors are then linearly projected into a lower-dimensional "patch embedding" (a token).
 - **Example:** A 256x256 image split into 16x16 patches gives $(256/16) * (256/16) = 256$ tokens. This is a much more manageable sequence length.

- **Tokenization via VQ-VAE (The DALL-E / Parti Approach):**

- A separate model, called a **VQ-VAE** (Vector Quantized Variational Autoencoder), is first trained to compress the image.
- The encoder of the VQ-VAE converts the image into a grid of "latent codes."
- Each code is mapped to its closest entry in a learned "codebook" (a dictionary of visual tokens).
- The image is now represented as a grid of discrete tokens (e.g., $32 \times 32 = 1024$ tokens).
- The Transformer learns to predict the sequence of these discrete tokens.

- **Stage 2: The Transformer - Learning and Generating**
- Once the image is a sequence of tokens, the process becomes very similar to training a language model like GPT.
- **Training:**
 - An image is tokenized into a sequence: [token_1, token_2, ..., token_N].
 - The tokens are fed into the Transformer in an autoregressive manner.
 - The model's task is to predict the *next token* in the sequence given all previous ones. The loss is calculated based on how accurately it predicts the next token.
 - Through this process, the Transformer learns the complex visual "grammar" and relationships between patches/tokens in the dataset.
- **Generation (Sampling):**
 - You start with a prompt (text or a start-of-sequence token).
 - The model predicts a probability distribution over what the first token should be.
 - You sample a token from this distribution (e.g., the most likely one or a random one based on probability).
 - You feed this generated token back into the model to predict the next token.
 - You repeat this process auto-regressively until the entire sequence of tokens is generated.
 - Finally, the sequence of tokens is converted back into an image using a decoder (e.g., the decoder from the VQ-VAE for methods using that approach).

Advantages of Transformer-Based Generation

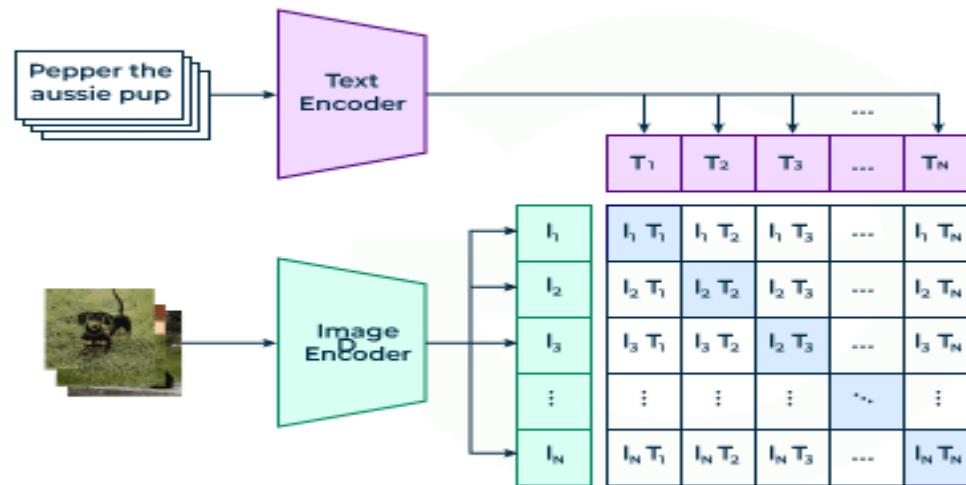
- **Exceptional Coherence and Long-Range Structure:** The self-attention mechanism allows every part of the image to directly interact with every other part. This is superb for maintaining global consistency (e.g., ensuring a creature has the same number of legs throughout the image).
- **Leverages NLP Advances:** They benefit directly from the massive scaling, architectures (e.g., Mixture-of-Experts), and training techniques developed for large language models.
- **Strong Textual Understanding:** When trained on image-text pairs, they excel at understanding and interpreting complex prompts, capturing style, composition, and concepts.

CLIP(Contrastive Language-Image Pretraining)

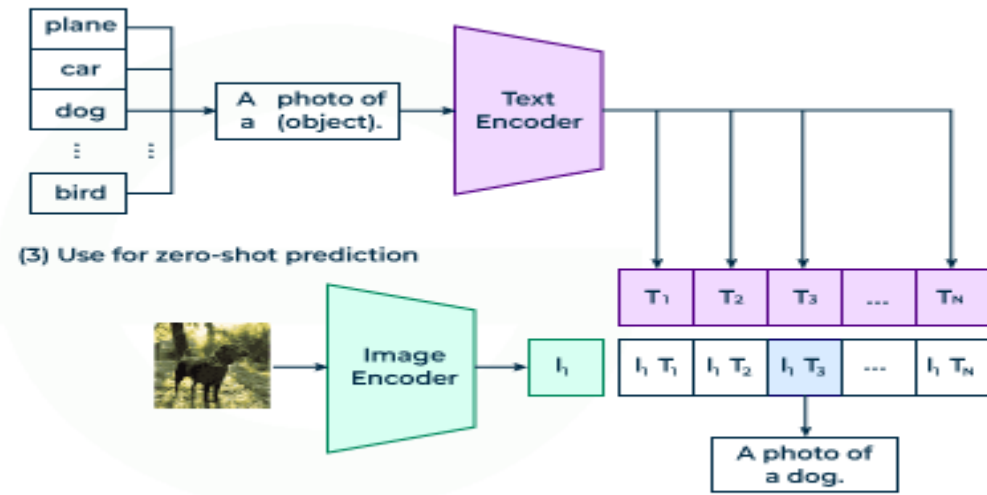
- **CLIP or Contrastive Language-Image Pretraining is an advanced AI model developed by OpenAI and UC Berkeley.**
- It has the unique ability to understand and relate both textual descriptions and images.
- It uses a novel training method that contrasts pairs of images and text which makes it highly useful tool for various real-world applications. In this article, we'll see the fundamentals of how CLIP works, its innovative approach and its other core concepts.

- This means that once trained, it can classify images into any category without needing to have been explicitly trained on that category. It is done by using both images and text in training, it can classify images into categories it wasn't explicitly trained on

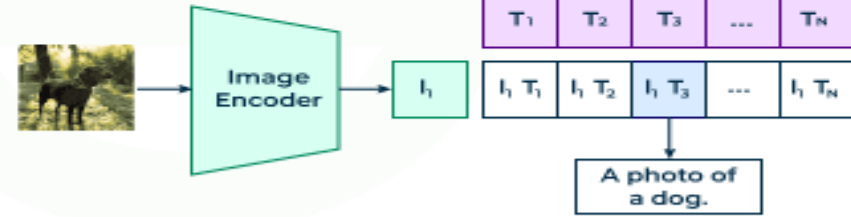
(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



- **1. Text Encoder**

- CLIP uses a Transformer-based model (similar to the model in the Attention is All You Need paper). This model converts text into embeddings, dense vectors that capture the meaning of the text. Its text encoder is a 63M-parameter model with 12 layers and 8 attention heads.

- **2. Image Encoder**

- For the image encoder, it experimented with both [ResNet](#) and [Vision Transformers \(ViT\)](#). At last ViT was chosen due to its superior performance in processing images. This encoder transforms images into embeddings that capture the image's key features.

- **3. Dataset**

- CLIP was trained on a massive dataset of 400 million image-text pairs sourced from the web. The team focused on using words that appeared at least 100 times in the English Wikipedia which ensures that 500,000 words were covered. This dataset called **WebImageText (WIT)** is important to CLIP's ability to generalize to various visual and textual concepts.

- **4. Training Objective**

- CLIP's goal is to align text and image embeddings which ensures that correct pairs of image and text are similar while incorrect pairs are not.

- **Cosine Similarity:** It maximizes similarity for matching pairs and minimizes it for non-matching pairs.
- **Training from Scratch:** Both image and text encoders are trained from the ground up, without pre-trained weights.
- **Projection:** The embeddings generated by the image and text encoder are projected into a shared space with the same dimensionality to allow for effective comparison.
- **Contrastive Learning:** During training, it distinguishes between correct and incorrect image-text pairs which optimizes the model to correctly identify matches.
- **Loss Function:** Cross-entropy loss is used to adjust the model, maximizing similarity for correct pairs and minimizing it for incorrect ones.
- **Inference:** After training, it calculates similarity scores between image-text pairs to determine relevance.

CLIP's Unique Features

- CLIP stands out from traditional models because of these key features:
- **Multimodal Training:** Traditional models process images and text separately. It, however, trains on both images and text at the same time which helps it to learn their relationship.
- **Zero-Shot Learning:** CLIP doesn't need to be retrained for new categories. After the initial training, it can classify images into any category even ones it hasn't seen before. This zero-shot learning is useful when retraining is not practical.
- **Self-Supervised Learning:** CLIP doesn't need explicit labels for every image or category. It learns by distinguishing between correct and incorrect image-text pairs which makes it a self-supervised model.

Real-World Applications of CLIP

- **Image Generation:** Models like [DALL·E 3](#) and [MidJourney](#) use CLIP to generate images based on text descriptions. It helps to translate text into image embeddings which ensures the images match the descriptions.
- **Image Segmentation:** SAM (Segment Anything Model) by Meta uses CLIP to understand text prompts and perform image segmentation which makes it easier to edit and manipulate images.
- **Content Moderation:** Social media platforms use CLIP to detect harmful or inappropriate content. It compares images to text descriptions to identify content that violates community guidelines.
- **Semantic Search:** It allows for text-to-image and image-to-text search. By turning both text and images into embeddings, it can match them accurately which provides better search results.
- **Visual Question Answering (VQA):** It can answer questions about an image's content based on a natural language query. For example, if you ask "What color is the car in this image?" it can give us a relevant answer.

Visual Transformers ViT- Dall-E2 and Dall-E3

- [Convolutional neural networks \(CNNs\)](#) have been at the forefront of the revolutionary progress in [image recognition](#) in the last ten years. Nonetheless, the field has been transformed by the introduction of [Vision Transformers \(ViT\)](#) which have implemented transformer architecture principles with image data.
- ViTs have shown outstanding success in different image recognition tasks, offering a new viewpoint on the processing of visual information.

Understanding the Architecture of Vision Transformers

- The core idea behind Vision Transformers (ViTs) is to treat images as sequences, similar to how words are treated in natural language processing (NLP). This innovative approach allows for the application of transformer architectures to image recognition tasks, fundamentally changing how visual data is processed.. The structure is comprised of a number of essential elements:
- **1. Image Patching**
- [Image Patching](#) is the initial step in the Vision Transformer process. This involves dividing images into smaller patches of a predetermined size. For example, a 224x224 pixel image can be segmented into **16x16 pixel patches**, resulting in **196 patches**. Each patch is then flattened into a vector, enabling the model to work with these smaller, manageable pieces of the image.
- **2. Positional Encoding**
- To maintain the **positional information** of the patches, **positional encodings** are added to the patch embeddings. This crucial step ensures that the model understands where each patch is located in the original image, allowing it to capture spatial relationships effectively.

- **3. Multi-Layer Transformer Encoder**

- The heart of the Vision Transformer is its **multi-layer transformer encoder**. This structure consists of:

- **Self-Attention Layers:** These layers allow the model to evaluate the relationships between different patches, helping it to understand how they interact with one another.
- **Feed-Forward Layers:** These layers apply non-linear transformations to the output of the self-attention mechanism, enhancing the model's ability to capture complex patterns in the data.

- **4. Classification Head**

- The **classification head** is a critical component of ViTs, utilized to generate predictions for image recognition tasks. A special token, often referred to as the **classification token (CLS)**, consolidates information from all patches, producing the final predictions. This aggregation of data ensures that the model leverages insights from the entire image rather than isolated patches.

How Vision Transformers Work?

- Vision Transformers begin by dividing an image into smaller, fixed-size patches. Each patch is then processed individually as part of a sequence, allowing the model to analyze the entire image through its components.
- The [self-attention mechanism](#) is fundamental to how ViTs operate. This mechanism allows each patch to influence the representation of other patches. Specifically, it computes attention scores that determine how much focus each patch should have on every other patch.
- This ability to weigh the importance of different patches enables Vision Transformers to understand **complex connections** and **interdependencies** throughout the entire image. As a result, ViTs can create more comprehensive and nuanced feature representations, capturing intricate patterns that might be missed by traditional convolutional networks.

- The training process for Vision Transformers involves adjusting the model's parameters to minimize the **prediction error** on labeled datasets. This is similar to the training process of other neural network architectures, where:
- Loss Function: A loss function is defined to quantify the difference between the predicted outputs and the actual labels.
- Backpropagation: The model uses backpropagation to update its weights based on the calculated loss, refining its ability to make accurate predictions.
- Optimization: Various optimization algorithms (e.g., Adam, SGD) are employed to enhance the learning process, ensuring that the model converges effectively.

Applications of Vision Transformers in Real-World Scenarios

- **Image Classification:** ViTs have demonstrated superior performance on standard datasets like ImageNet, establishing their effectiveness for various image classification assignments.
- **Object Detection:** ViTs outperform in object detection due to their capability to capture global contexts, which is essential for tasks such as autonomous driving and surveillance.
- **Semantic Segmentation:** ViTs show potential in medical applications when used for pixel-level classification tasks.
- Recent studies are investigating the utilization of ViTs in generative models for the development of new applications in art and design.

Conclusion

- Vision Transformers are changing the way image recognition works, disrupting the traditional reign of convolutional neural networks. Their distinctive structure, which relies on self-attention mechanisms, enables the detection of intricate patterns and relationships in images.
- Despite facing obstacles, Vision Transformers are seen as a crucial technology for the future of computer vision due to their advantages and adaptability. With ongoing research development, we anticipate further creative uses and enhancements in the capabilities of Vision Transformers for image recognition.

Advantages and Disadvantages of Vision Transformers Over CNNs

- **Global Context Awareness:** In contrast to CNNs that prioritize local features, ViTs are able to understand the global relationships within an image, resulting in better performance for challenging image recognition assignments.
- **Scalability:** Vision Transformers perform efficiently with data, frequently surpassing CNNs when trained on extensive datasets, as they can efficiently make use of additional computational resources.
- **Versatility:** ViTs can adapt to different applications due to their flexibility in handling varying input sizes and lack of rigid architectures.
- **Transfer Learning:** ViTs demonstrate proficiency in transfer learning by utilizing knowledge from pre-existing models to achieve high performance on similar tasks with minimal labeled data available.

GPT-4V

- **What is GPT-4 Vision?**
- GPT-4 Vision (GPT-4V) is a multimodal model that allows a user to upload an image as input and engage in a conversation with the model. The conversation could comprise questions or instructions in the form of a prompt, directing the model to perform tasks based on the input provided in the form of an image.
- The GPT-4V model is built upon the existing capabilities of GPT-4, offering visual analysis in addition to the existing text interaction features.
- We have covered the [beginner-friendly introduction to the OpenAI API](#) that'll help you get up to speed on the developments prior to the release of the GPT-4V model.

Key Capabilities of GPT-4 Vision

- **Visual inputs:** The key feature of the newly released GPT-4 Vision is that it can now accept visual content such as photographs, screenshots, and documents and perform a variety of tasks.
- **Object detection and analysis:** The model can identify and provide information about objects within images.
- **Data analysis:** GPT-4 Vision is proficient in interpreting and analyzing data presented in visual formats such as graphs, charts, and other data visualizations.
- **Text deciphering:** The model is able to read and interpret handwritten notes and text within images.

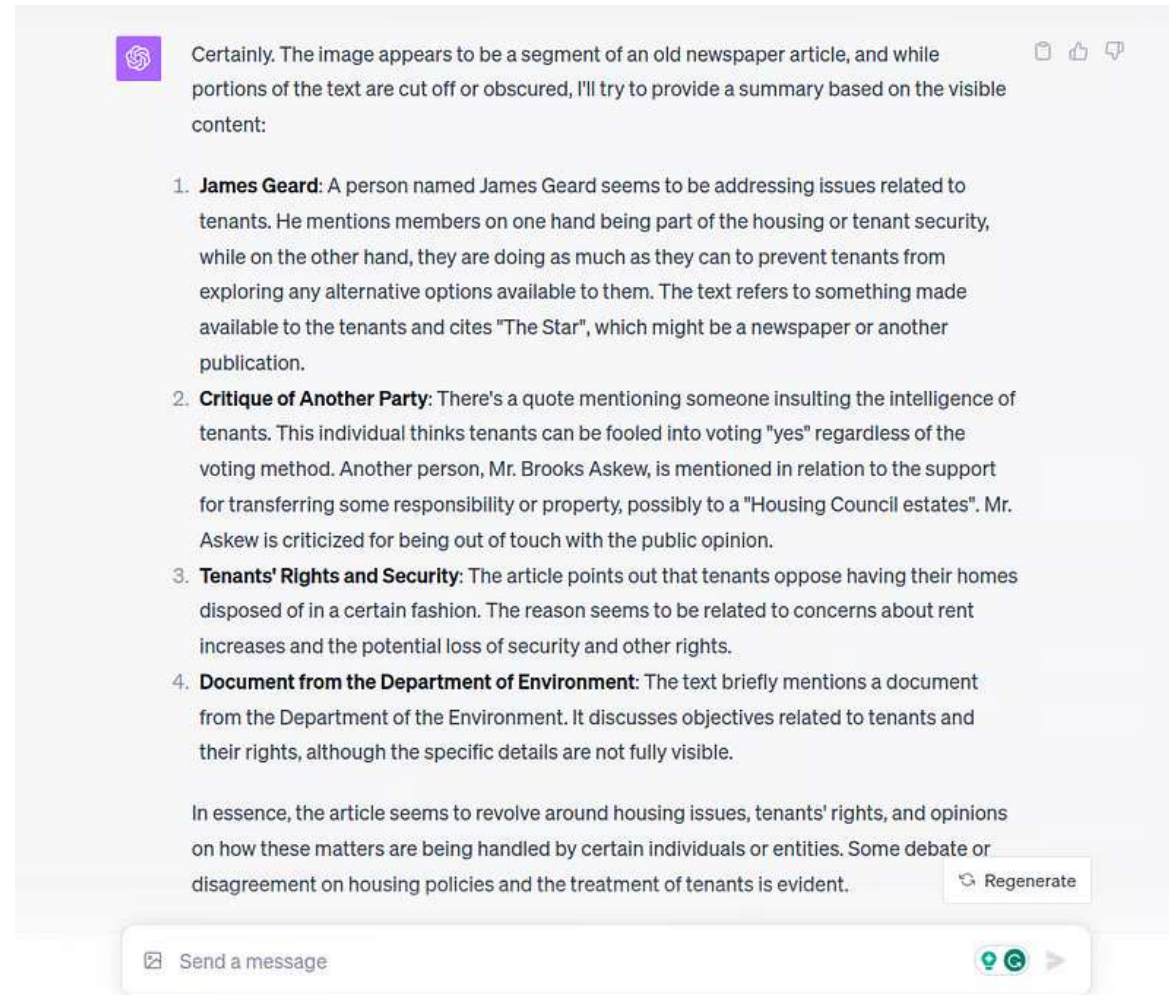
Hands-On: Getting Started with GPT-4 Vision

- GPT-4 Vision is currently (as of Oct 2023) available for the ChatGPT Plus and Enterprise users only. ChatGPT Plus costs \$20/month, which can be upgraded to from your regular free ChatGPT accounts.
- Assuming you're completely new to ChatGPT, here's how to access GPT-4 Vision:
- Visit the [OpenAI ChatGPT](https://openai.com/chatgpt) website and sign up for an account.
- Login to your account and navigate to the "Upgrade to Plus" option.
- Follow through with the upgrade to gain access to ChatGPT Plus (Note: this is a monthly subscription of \$20)

- **1. Academic research**
- GPT-4 Vision's integration of advanced language modeling with visual capabilities opens up new possibilities in academic fields, particularly in deciphering historical manuscripts. This task has traditionally been a meticulous and time-consuming endeavor carried out by skilled paleographers and historians.
- We first give an image that appears to be part of an old newspaper article:



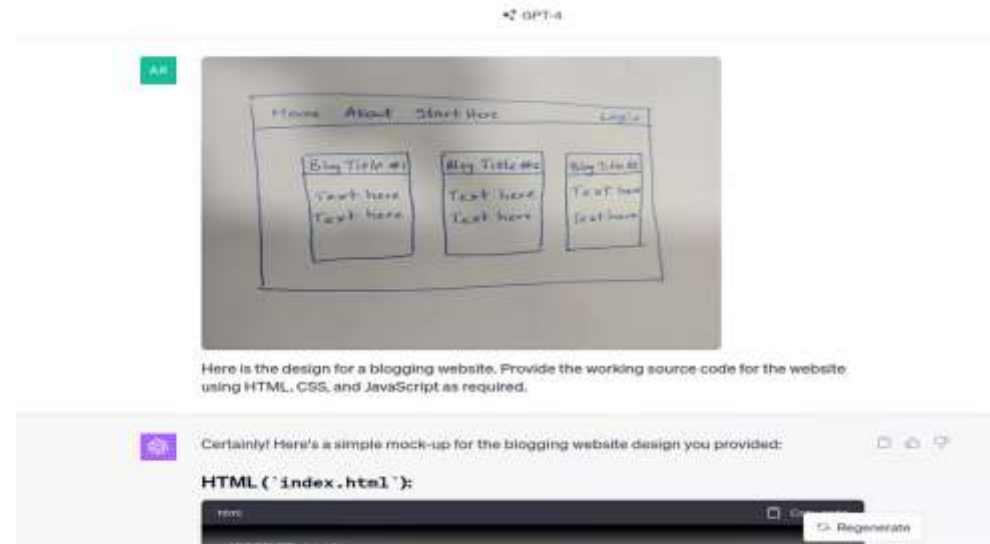
- GPT-4 Vision does a good job of reading the contents of the image and interpreting it:



2. Web development

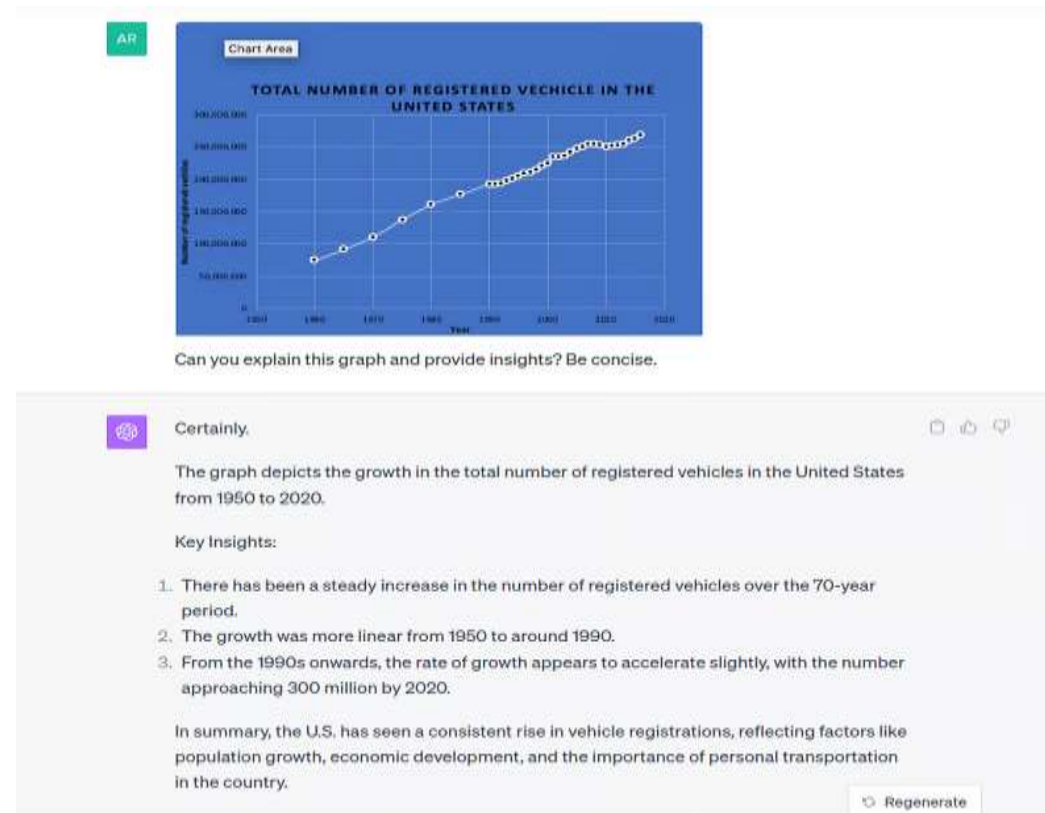
- The GPT-4 vision can write code for a website when provided with a visual image of the design required. It takes from a visual design to source code for a website. This single ability of the model can drastically reduce the time taken to build websites.
- Let us prompt the GPT-4 Vision with a hand-drawn simple design for a blogging website.

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3. Data interpretation

- The model is capable of analyzing data visualizations to interpret the underlying data and provide key insights based on the visualizations. To test out this feature, we can simply give a plot and ask for insights.



4. Creative content creation

- With the advent of ChatGPT, social media is full of various prompt engineering techniques, and many have found surprising, creative ways to use the generative technology to their advantage.
- For this tutorial, we will use the DALL-E-3 (which is also available in ChatGPT Plus) together with GPT-4 Vision, to creatively create a social media post.
- **Step 1:** Ask GPT-4 to create a prompt to generate an image. Let's say you want to create a post contrasting the differences between a data scientist role in a startup vs a corporate one.

Issues of Image Generation models

- **Bias and Discrimination**
- **Reinforcing Stereotypes:**
 - Models trained on biased datasets can perpetuate harmful stereotypes, such as showing over-reliance on Western beauty standards or misrepresenting specific demographics.
- **Ageism:**
 - AI may fail to represent people of all ages, often producing images of younger individuals and overlooking or misrepresenting older adults, reinforcing negative stereotypes about age and activity.

- **Ethical and Legal Issues**
- **Copyright Infringement:**
 - AI models are often trained on vast amounts of images, some of which may be copyrighted, raising questions about intellectual property and attribution.
- **Non-Consensual Likeness:**
 - Models can generate realistic images of people without their permission, creating potential for misuse and raising significant ethical and legal questions about control over one's image.
- **Misinformation and Deepfakes:**
 - The ability to generate highly realistic images can be exploited to create convincing fake content for misinformation campaigns or phishing scams.

- **Technical Limitations**
- **Lack of True Understanding:**
- Models lack a deep understanding of the real world, focusing on shapes and colours rather than the semantic relationships and context that characterize authentic images.
- **Spatial Inconsistencies:**
- Generated images can exhibit flaws in spatial reasoning, such as unnatural object placement, inconsistent background elements, or failure to correctly represent relative positions and occlusions.
- **Incomplete Scene Composition:**
- Models may struggle to incorporate multiple complex components into a single image, leading to a decline in overall image quality and coherence.

- **Privacy and Security**

- **Privacy Violations:**

- The ability to generate images resembling real individuals without consent directly infringes on privacy rights and can be used in deceptive ways.

- **Cybersecurity Risks:**

- AI-generated images can be used by malicious actors to create fake identities for phishing attacks or other social engineering schemes.

- **Other Issues**
- **Lack of Originality:**
 - Because AI models rely on existing training data, their output can sometimes be generic, lacking true creativity or groundbreaking artistic merit.
- **High Computational Costs:**
 - Training and running advanced image generation models requires significant computational resources and energy, presenting accessibility challenges and contributing to a large carbon footprint.